

# RetailRadar

Detecting the **start** and the **end** of retail euphoria from crowd data alone

A walk-forward study on nine years of social-media attention

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Last change: every frame fitted by measurement rather than by eye — 33 overfull boxes to zero at 77 pages; two silent content losses repaired (the hypotheses slide was pushing its own SO WHAT strip off the page, and \stat counters used inside a column were collapsing to 48pt slivers); dashboard screenshots roughly doubled in rendered area by turning those frames sideways and splitting the paired tab slides one-per-slide; long paths made breakable and prose table columns given declared widths

# What this deck claims, in one slide

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333

price-defined boom-bust  
episodes, 2017–2026

59

instruments (34 theme  
ETFs, 25 data-chosen  
singles)

8 days

median warning before the  
peak (GET OUT)

0.20

false alarms per  
instrument-year (GET  
OUT)

- **The system works as risk timing, not as alpha.** The END detector flags roughly one in five detectable manias a median eight days *before* the peak. A mechanical buy/sell translation was tested under a pre-stated criterion and **rejected** — and that rejection is in this deck, not buried.
- **The strongest single result is a state, not an alert.** When the crowd is  $\geq 2\times$  its own normal *and* price is already in a boom, a  $\geq 10\%$ -in-7-days drop begins within 30 days **62% of the time, versus 19% on ordinary days** (90% CI on the uplift [+28pp, +50pp]).
- **Nothing here is a tuned threshold.** Every constant is LEARNED, DERIVED from an already-validated constant, a stated CONVENTION with an ablation, GROUND TRUTH, or a recorded DESK DECISION.

# How this deck is ordered — and why

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## Part I — the answer

1. Aims and hypothesis
2. What the desk sees (dashboard)
3. Results, quantitative

## Part II — the defence

4. Data and ingestion
5. Ground truth: writing the exam
6. Features and the importance battery
7. The tournament and the walk-forward record
8. Influence: the companion graph study
9. Limitations, and what is next

**Why results before method.** A trading audience should be able to leave after slide 12 knowing exactly what the system does, what it does not do, and how often it is wrong. The method sections exist to survive the question *“why should I believe that number?”* — which is a different meeting from *“what is the number?”*

**Reading the colours.** **Teal** is always GET IN (euphoria starting), **brick** is always GET OUT (euphoria ending), on every chart in the deck and on every screen in the product.

## Aims — what a PM actually needs

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- **The problem.** Retail attention episodes — GameStop 2021, gold 2026, the meme complex — follow a recognisable arc: a crowd arrives, sentiment goes one-way, attention grows super-exponentially, price peaks, and a bust of 15–90% follows within weeks.
- **What the literature already gives us.** Attention extremes predict *reversal*, not continuation (Barber & Odean 2008; Barber et al. 2022). That is a correlation.
- **What a desk needs instead is a clock.** Has an episode *started* (the rally has fuel left), and is it *ending* (the top is near)? Two different questions, two different observable signatures.

**SO WHAT** The deliverable is not a return forecast. It is a two-state warning a PM can read in ten seconds: **GET IN** — a crowd episode is starting on this name; **GET OUT** — the one we are already in is ending.

# Hypotheses, stated before the tests were run

|    | Hypothesis                                                                                                            | Verdict (evidence)                                                                                                      |
|----|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| H1 | The START of a euphoria episode is detectable from crowd data alone, above a random floor.                            | <b>yes</b> — weakly. 23% of detectable starts, AP 0.098 vs 0.072 floor.                                                 |
| H2 | The END is detectable from crowd data alone with usable warning.                                                      | <b>yes</b> — 16/122 detectable tops, median 6-day lead, crowd-only.                                                     |
| H3 | A learned model (LR / GBM / MLP) beats the rule bank.                                                                 | <b>no</b> — every learner ties inside bootstrap noise or falls below random. Rules win by parsimony.                    |
| H4 | The alerts translate into a mechanical trade (onset→buy, top→sell) over 20 days.                                      | <b>no</b> — <b>rejected</b> under its own pre-stated criterion.                                                         |
| H5 | Adding price as a <i>gate</i> (not a feature) improves the END detector.                                              | <b>yes</b> — capture 16→26 on matched years, AP 0.286→0.435, CI [+3.5, +13]pp.                                          |
| H6 | High-predictive authors can be identified from behaviour and reply-graph position before their track record is known. | <b>no</b> — real on a random split ( $p = 0.005$ ), but <i>nil</i> on unseen authors, which is the deployment question. |

**SO WHAT** Three of the six hypotheses failed. They are on the same slide as the successes because a defence is won by the results you volunteer, not the ones you survive.

# The one rule the whole project is built on

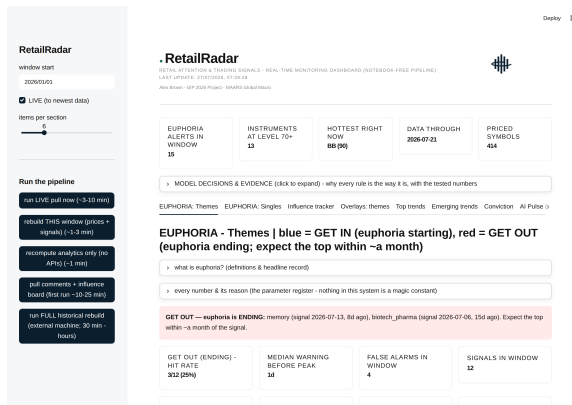
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**The crowd-only rule.** Every *predictive* input is built from social-media aggregates. Price never enters any feature, level, gate or alert in the headline detectors. Price exists only to define ground truth and to score the detectors.

A unit test enforces it: no detector function accepts a price argument.

- **Why bother.** The claim being defended is “*the crowd alone called it*”. The moment price leaks into a feature, that claim is unfalsifiable and the study is worth nothing.
- **The recorded exception.** A *second* signal family (2026-07-24 desk decision) permits price as a **gate** — a prerequisite, never a score input. It is reported separately, and the crowd-only detectors remain the research headline.

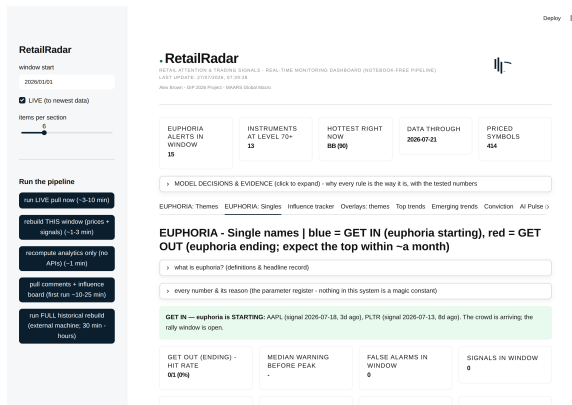
# The product: two signals, one chart



The euphoria pane. One line per name, GET IN and GET OUT marked as vertical rules — deliberately not shaded bands: a shaded region invites the reader to believe the whole region was called, when what was actually measured is a dated alert.

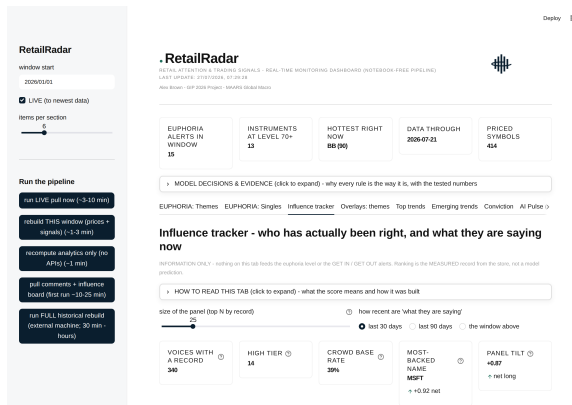
**SO WHAT** The chart shows conclusions only. Every ablation, tournament and confidence interval lives in the notebooks; the screen carries the decision.

# The signal, close up



Singles pane. The scorecard above the chart recomputes hit rate, median lead and false-alarm count for whatever window is selected — judged by the same functions the research record uses, so the screen cannot flatter the study.

# Influence: what the names that matter are pushing



Direction  $\times$  backing. Horizontal = net direction of the calls, vertical = influence behind them, bubble size = number of calls, colour = sentiment. Top-right therefore reads as *crowded and bullish*.

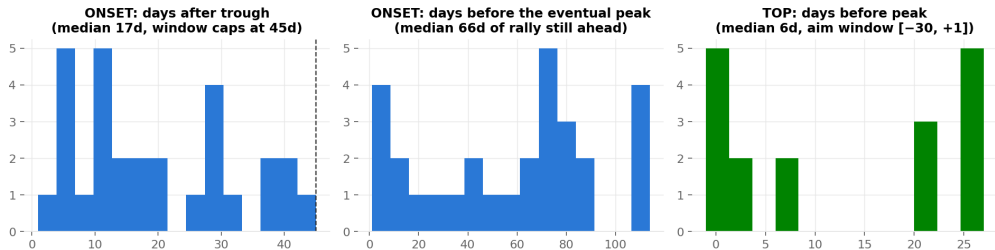
**SO WHAT** This pane is **information, not signal**. It does not enter the euphoria detector and is never scored as if it did.

## The walk-forward operating record

|                                         | Captured | Detectable | Rate  | False alarms | FA / instr-yr |
|-----------------------------------------|----------|------------|-------|--------------|---------------|
| GET OUT (shipped, boom-gated, smoothed) | 24       | 122        | 19.7% | 39           | 0.195         |
| GET IN (shipped, phase-aware, smoothed) | 20       | 125        | 16.0% | 124          | 0.25          |
| Crowd-only onset (research headline)    | 29       | 125        | 23.2% | 169          | 0.348         |
| Crowd-only top (research headline)      | 16       | 122        | 13.1% | 44           | 0.128         |

- **Lead times.** GET OUT fires a median **8 days** before the peak. GET IN fires a median **17 days after** the trough, with a median **66 days of rally still ahead**.
- **Two denominators, always.** 29 of 125 *detectable* starts is also 8.7% of all 333 episodes. Coverage is deep in 2020–22 and 2026 and thin to absent in 2023–25 — the detector is *blind* there, which is not the same as wrong.
- **One number above budget, printed on the pane itself.** Onset false alarms run 0.348/instrument-year against a derived budget of 0.23.

# How much warning, exactly?

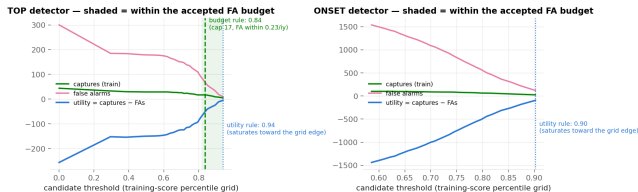


Every scored alert, as a distribution rather than as a mean. *Left:* GET IN fires a median 17 days after the low (the 45-day window is the hard cap). *Middle:* at that moment a median 66 days of rally is still ahead. *Right:* GET OUT fires a median 6 days before the peak, inside a  $[-30, +1]$  day aim window.

**SO WHAT** These are the two numbers a PM actually trades on. A GET IN with two months of rally still ahead is actionable; a GET OUT six days before the top is a *risk* instruction, not an entry — there is no time to build a short, but there is time to stop adding.

# Why 0.63 and not 0.60? — the threshold picks itself

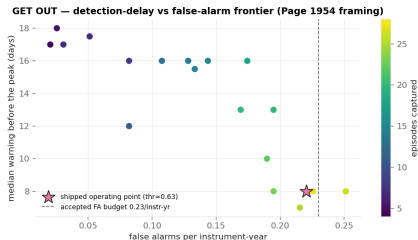
The training evidence behind each threshold — two selection rules, both drawn, neither hand-picked



The full candidate grid, for the TOP detector (left) and the START detector (right), drawn from *training* years only. Green = episodes captured, pink = false alarms, blue = utility (captures — false alarms). Shading marks the region inside the accepted false-alarm budget.

**SO WHAT** This is the answer to “*why 85, why not 80 or 90?*” — there is no hand-picked number. Two rules are stated in advance and both are drawn: the **budget rule** takes the highest-capture threshold that stays inside a derived false-alarm budget, and the **utility rule** maximises captures minus false alarms. Where they disagree, the tighter one ships. The chart is here so a reviewer can see the whole curve the choice was made from, not just the point.

# The shipped operating point, on its own frontier



Every threshold the GET OUT detector could have shipped at, plotted as warning-time against false-alarm rate and coloured by episodes captured. The star is what actually shipped ( $\approx 0.63$ ); the dashed line is the accepted budget of 0.23 false alarms per instrument-year.

**SO WHAT** The frontier is the honest picture of the trade: you can buy 18 days of warning instead of 8, but only by capturing five episodes instead of twenty. The shipped point sits at the capture-maximising end of what the budget allows — a *stated* preference for coverage over lead time, recorded as a desk decision rather than discovered by tuning.

# The strongest result: the danger state

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## 62%

of danger-state days are followed by a  $\geq 10\%$ -in-7-days drop within 30 days

## 19%

on ordinary candidate days

90% instrument-cluster CI on the uplift: [+28pp, +50pp]

**The danger state** is not a model. It is the conjunction of two conditions the system already computes:

- the crowd is  $\geq 2\times$  its own 120-day normal, *and*
- price is already in a boom off its own 120-day low.

A combined price+crowd *alert* model was also tested and did **not** clear its criterion — its 65% hit rate is indistinguishable from the 62% baseline of its own candidate days.

**Which is the finding:** the candidacy state carries the information, so the product ships the state as a standing amber band, and uses the END alerts to time the peak *within* it.

**SO WHAT** This is the slide to remember. The most useful thing the system produces is a *regime flag* that triples the base rate of a near-term double-digit drawdown.

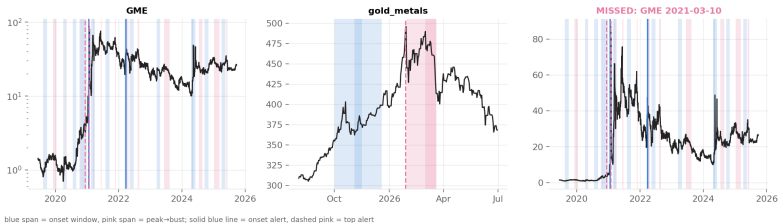
# Three episodes, and what the detector said



Price on a log axis. Blue span = the 45-day window in which a START would count as caught; the blue dot is the measured 120-day low, the pink dot the confirmed peak. *Left*: GameStop, low 2020-10-06, peak 2021-01-27. *Middle*: gold, low 2025-10-02, peak 2026-01-29, with the peak-to-bust span shaded. *Right*: energy 2020 — a V-recovery that qualifies as an episode by construction.

**SO WHAT** Energy is on this slide on purpose. It is the limitation made visible: a boom off a 120-day low admits a post-crash *rebound*, not only a mania. The crowd features have to earn that distinction, and no hand-written exclusion rule is used to hide the cases where they do not.

# The same names, with the alerts drawn on



Blue span = onset window, pink span = peak→bust, solid blue line = a START alert, dashed pink line = a TOP alert. *Left*: GameStop. *Middle*: gold 2026. *Right*: a labelled **miss** — GME 2021-03-10, the second squeeze, where no alert fired in the aim window.

**SO WHAT** The right-hand panel is the one to dwell on. The March 2021 re-squeeze was not caught, and the reason is diagnosable rather than mysterious: the crowd never re-quietened between the January peak and the March one, so “attention versus its own normal” was already elevated and had no room left to spike. Consecutive manias on the same name are the detector’s known blind spot, and it is a *stated* one.

# Gold 2026 and GameStop 2021: desk END vs crowd END



Dark red = the shipped boom-gated GET OUT; pink = the crowd-only research END; shaded = the actual peak-to-bust fall. *Left*: gold 2026 — one clean alert just before the January peak. *Right*: GameStop 2020–22, where the crowd-only detector fires repeatedly through the squeeze and the gated version fires far less often.

**SO WHAT** This is what the price *gate* buys, in one picture: the same crowd score, filtered to days when price is already in a boom, stops calling the top of a rally that has not started. It costs research purity — which is why the crowd-only detector remains the headline and the gated one is reported separately.

## What we tested and rejected

| Claim       | n   | Alert 20d | Baseline | Diff   | 90% CI           |
|-------------|-----|-----------|----------|--------|------------------|
| ONSET → BUY | 215 | +0.34%    | +0.98%   | −0.63% | [−1.51%, +0.26%] |
| TOP → SELL  | 67  | −0.21%    | +1.34%   | −1.55% | [−5.63%, +2.63%] |

- **Criterion, written before any return was computed:**  $n \geq 30$  judged alerts; mean 20-day forward return beats the same-instrument unconditional baseline in the expected direction; 90% instrument-cluster CI of the difference excludes zero.
- **Both verdicts: REJECTED.** The TOP→SELL direction is right, but at  $n = 67$  the interval spans zero. The ONSET→BUY edge is simply not there at 20 days.
- A short-after-GET-OUT overlay was also measured: annualised Sharpe  $-0.82$ . It is in the record for the same reason.

**SO WHAT** The detectors time *risk*. Avoiding the left tail of a long you already hold is a different and easier trade than timing a short entry — and it is the one the evidence supports.

## Performance battery at the true operating point

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|         | Precision | Recall | F1    | MCC   | AP    | Base rate |
|---------|-----------|--------|-------|-------|-------|-----------|
| GET OUT | 0.395     | 0.450  | 0.421 | 0.037 | 0.449 | 0.374     |
| GET IN  | 0.151     | 0.061  | 0.087 | 0.059 | 0.084 | 0.062     |

- **Why these metrics.** Precision–recall rather than ROC because positives are rare (Davis & Goadrich 2006); Matthews correlation because it is the one summary that cannot be gamed by class imbalance (Chicco & Jurman 2020).
- **Read AP against base rate, always.** GET OUT's 0.449 against a 0.374 base rate is a modest lift. GET IN's 0.084 against 0.062 is a smaller one. Neither is a headline; both are honest.
- **MCC near zero is the uncomfortable number** and it is here on purpose: at the operating point these are *sparse warnings*, not classifiers, and a metric built for balanced classification says so.

# Data — what goes in, and what never does

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## Crowd data (the only predictive input)

- 17 finance subreddits + X + StockTwits, 2017-01-01 → present.
- Daily, **text-free** aggregates: per-name mention counts, post counts, average sentiment, net-bullish share.
- Sentiment is VADER-scored *at ingestion* — text is scored and discarded at the boundary.

## Prices (scoring only)

- Bloomberg daily closes, ~287 symbols, append-only store.

### The privacy contract, enforced in code.

No titles, bodies, author names or post IDs ever cross the ingestion boundary. The influence store is pseudonymous and text-free, and a write-time check *raises* if any text column reaches it.

This is not decoration: it is what makes the store shareable, and it is checked on every write rather than trusted to reviewer discipline.

**SO WHAT** Every number downstream inherits one weakness — archive depth. It is deep in 2020–22 and 2026, thin in 2023–25. That is reported on every rate rather than averaged away.

# The universe, and why it is data-chosen

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- **59 instruments:** 34 theme-anchor ETFs (rates/bonds and real estate excluded by desk decision) and the **top-25 most-mentioned single names** with  $\geq 3,000$  scored posts.
- **Data-chosen, not hand-picked** — because the point is catching the *next* GameStop. A universe curated with hindsight would guarantee a flattering backtest and no forward value.
- **The panel self-expands where the crowd points.** A monthly review mines the panel for `r/<name>` referrals, qualifies a candidate at  $\geq 100$  unique referring authors per 28 days (the existing coverage floor, reused — not a new number), screens it for finance content, and admits at most one community per month into an EXPLORATION tier. The founding 17 are a frozen CORE.

**SO WHAT** Every panel change steps the mention-share denominator, so every addition is written to a committed manifest. This is the `source_breadth` lesson institutionalised: coverage-regime changes must be recorded and measurable, never silent.

# Writing the exam before sitting it (Notebook 01)

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**An episode** is one complete boom-then-bust arc: the low a rally started from, the confirmed peak, and the fall that followed.

- Up 25% (theme) or 50% (single) off the own 120-day low. . .
- . . .then down 15% / 30% within 90 days.
- Singles are held to the harder bar because they move further on nothing.

**Defined from price only.** No crowd feature is ever allowed to influence what counts as an episode — otherwise the exam is written by the candidate.

**SO WHAT** *Detectable* and *caught* are different columns and are never merged. An episode in a thin archive is one the detector was blind to — counting it as a miss would understate the system; counting it as absent would flatter it.

333

episodes, 2017–2026

136

gradeable on the *top*

126

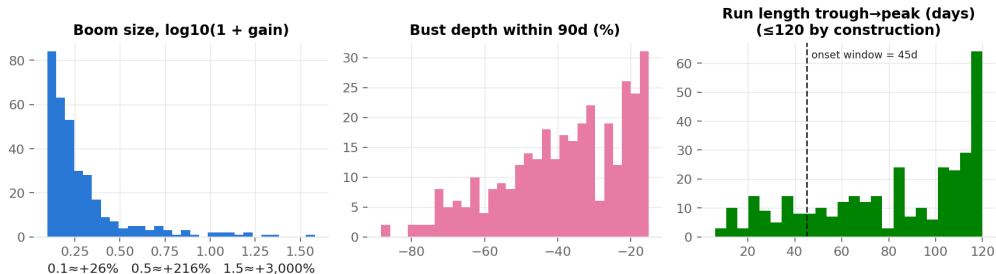
gradeable on the *start*

86 d

median run from low to peak

Median boom +63.5%, median bust  
–36.3%.

# The distributions the detector has to live with

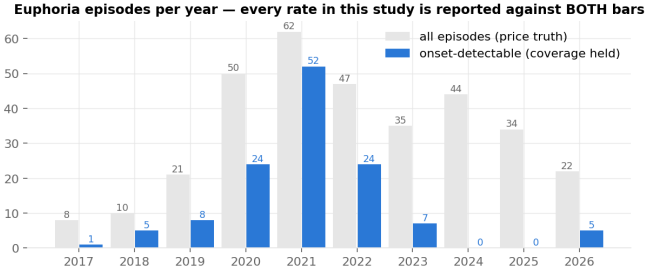


The price-only catalog, three panels. *Left*: boom size on a log scale, because gains run from +25% to +3,000% and a linear axis would show one bar. *Middle*: bust depth within 90 days. *Right*: run length low→peak, with the 45-day onset window marked.

**SO WHAT** Run lengths cap at 120 days *by construction* — the trough is a measured 120-day low. Slow multi-quarter rallies therefore read as starting later than their narrative start, and that is carried as a stated limitation rather than hidden.

# The coverage problem, stated plainly

| Year               | 2017 | 2018 | 2019 | 2020 | 2021      | 2022 | 2023 | 2024     | 2025     | 2026 |
|--------------------|------|------|------|------|-----------|------|------|----------|----------|------|
| Episodes           | 8    | 10   | 21   | 50   | 62        | 47   | 35   | 44       | 34       | 22   |
| Gradeable on start | 1    | 5    | 8    | 24   | <b>52</b> | 24   | 7    | <b>0</b> | <b>0</b> | 5    |
| Gradeable on top   | 1    | 5    | 8    | 24   | <b>58</b> | 22   | 5    | <b>0</b> | <b>0</b> | 13   |



Grey = every episode price found. Blue = the subset the archive can actually grade. The gap *is* the coverage problem, and every rate in the study is reported against both bars rather than against whichever one flatters it.

# IF ASKED: “41% of your evidence is one year”

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**The challenge is fair.** 2021 supplies 52 of the 126 gradeable starts and 58 of the 136 gradeable tops. 2024 and 2025 supply **zero**.

**The honest answer is *partially* — with three mitigations, none of which is a re-weighting:**

- Walk-forward: 2021 is *out-of-sample* on the year it is scored, fitted only on 2018–20.
- Label sensitivity: capture is flat at 22.4% / 23.2% / 22.1% under softened / standard / hardened episode definitions — the result is not an artefact of where the boom-bust line was drawn.
- 2026 is a fresh, independent, coverage-rich year and is in the record on its own: 13 gradeable tops.

## Why not just collect more data?

Because the missing years are missing from the *archive*, not from our pull. Reddit's public history thins sharply for 2023–25 in the submission endpoints available to us. Folding comment bodies into the aggregates is the one lever that could recover them, and it is the first item on the future-work slide for exactly that reason.

**What we refuse to do** is average across years and quote a single blended rate. A blended rate hides that the detector was never *tested* in 2024–25 — it presents blindness as accuracy.

## Two denominators, reconciled

|                        | Catalog | Scored | Why they differ                                                    |
|------------------------|---------|--------|--------------------------------------------------------------------|
| Gradeable on the START | 126     | 125    | 2017 has no prior year to fit on, so its 1 episode is never tested |
| Gradeable on the TOP   | 136     | 122    | TOP scoring starts 2020; the 14 tops of 2017–19 are never tested   |

- **Why this slide exists.** The deck quotes 126 and 136 when describing the *catalog* and 125 and 122 when quoting a *capture rate*. That looks like an inconsistency, so here is the arithmetic:  $126 - 1 = 125$  and  $136 - 14 = 122$ , exactly.
- **The rule that produces it.** A walk-forward study can only score a year it has a training history for. The unscored episodes are not discarded from the catalog and not counted as misses — they are outside the test set, and saying so is cheaper than being caught.

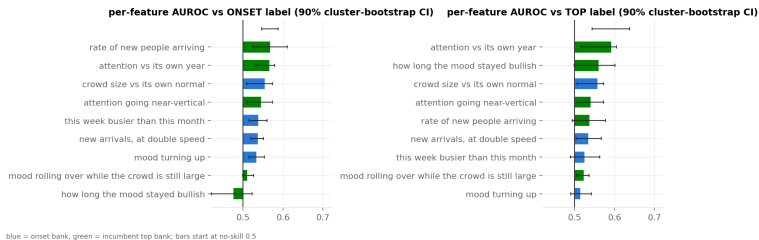
**SO WHAT** Every denominator in this deck is traceable to a line in `docs/research/nb01_episode_stats.json` or `nb04_final_eval.json`. None is rounded to make a rate look better.

# Which crowd measurements are worth anything? (Notebook 02)

| Feature (plain English)          | What it measures                    | Where its numbers came from                         |
|----------------------------------|-------------------------------------|-----------------------------------------------------|
| this week busier than this month | 7d share — 28d share, ranked        | DERIVED — the project’s existing roll windows       |
| crowd size vs its own normal     | 7d share ÷ own 120d median          | DERIVED — the hype gate, made continuous            |
| mood turning up                  | 14d change in 14d net-bullish share | DERIVED — the fade rule, pointed the other way      |
| new arrivals, at double speed    | 14d change in mention share         | DERIVED — the influx feature at half the horizon    |
| attention going near-vertical    | slope of attention                  | CONVENTION — identical to E5, reported as identical |

**SO WHAT** Not one window in the bank is new. Every one traces to a constant that already existed in the validated detector — so no reviewer can point at a window and ask “why that one?”

# No single feature is a detector — and that is the finding



Per-feature AUROC against the START label (left) and the TOP label (right), 90% instrument-cluster bootstrap intervals. Bars start at 0.50 — the coin-flip line — because distance from chance, not absolute height, is the quantity of interest. Colour marks which bank the feature sits in: blue the START bank, green the incumbent TOP bank.

**SO WHAT** AUROCs sit in **0.51–0.59**. Eight of ten feature/label pairs clear no-skill with the interval excluding 0.5, and precision runs  $1.2\text{--}1.5\times$  base rate. Weak-but-real is exactly the raw material a combination step is for — and anyone shipping a single magic crowd variable would be shipping an overfit.

# The feature we rejected *because* it scored best

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`source_breadth` — how many platforms are talking — posted the **best raw AUROC of any candidate**.

It is **rejected**.

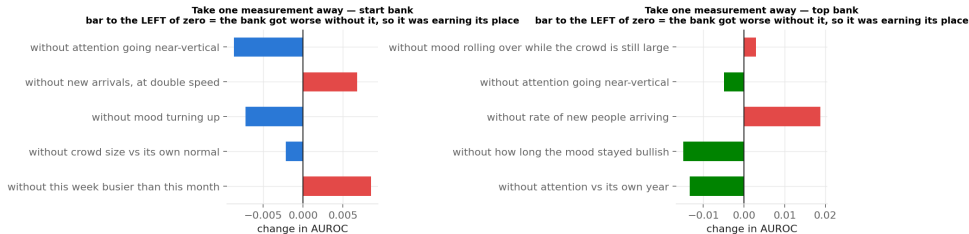
The archive only gained its second and third source in 2026. Before then the feature is a constant; afterwards it mostly encodes *which year it is*. Its skill is a coverage-regime artefact.

**IF ASKED** — “you rejected your best feature; isn’t that just cherry-picking in reverse?”

The rejection rule was written before the scores were computed: a feature must be measurable across the whole sample, not only after a data-collection change. Scoring well for the wrong reason is a *research* false positive, and catching it here costs a slide — catching it in production costs the desk’s trust.

**SO WHAT** The single most persuasive thing in this deck is a feature that is not in the model.

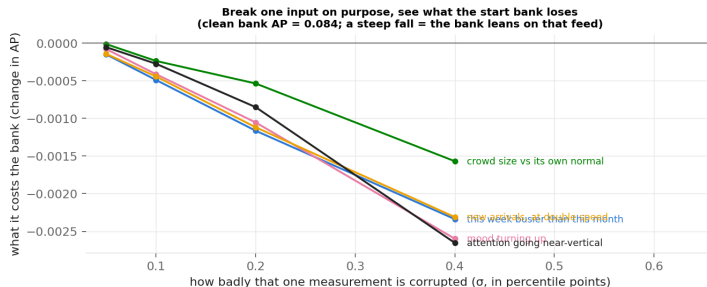
# Is the bank one good feature in disguise?



Drop-one ablation of the equal-weight bank scorer. Bars to the left of zero mean removal *hurt*. Equal weights mean nothing is fitted here, so a delta cannot be a re-optimised weight quietly compensating.

**SO WHAT** No. Every bar is inside the noise band — the bank is a committee, not a soloist plus four passengers. That is a **stability** argument: a bank where one input carried everything would break the day that one data source changed.

# If a data feed degrades, does the signal collapse?



Perturbation test adapted from the thesis's graph-corruption study: add increasing rank-space noise to one feature at a time (mean of 20 draws) and watch what the bank loses.

**SO WHAT** No cliff inside the tested range, and  $\sigma = 0.40$  is closer to *destroyed* than to *degraded*. A bad scrape day degrades the signal rather than inverting it — which is the property a live system needs and a backtest never checks.

## The tournament — criterion written before results

| Task  | Model        | AP           | AUROC | Captured | FA/instr-yr |
|-------|--------------|--------------|-------|----------|-------------|
| Onset | <b>rules</b> | <b>0.098</b> | 0.565 | 29/125   | 0.35        |
| Onset | gbm          | 0.098        | 0.609 | 26/125   | 0.37        |
| Onset | logreg       | 0.089        | 0.595 | 28/125   | 0.36        |
| Onset | random       | 0.072        | 0.504 | 43/125   | <b>1.04</b> |
| Onset | mlp          | 0.063        | 0.462 | 31/125   | 0.49        |
| Top   | <b>rules</b> | <b>0.286</b> | 0.557 | 16/122   | 0.13        |
| Top   | logreg       | 0.265        | 0.592 | 16/122   | 0.15        |
| Top   | mlp          | 0.228        | 0.482 | 10/122   | 0.13        |
| Top   | gbm          | 0.213        | 0.525 | 12/122   | 0.16        |
| Top   | random       | 0.204        | 0.486 | 10/122   | 0.15        |

- **ONSET** — GBM ties rules on AP but the paired bootstrap CI of the difference straddles zero  $[-0.017, +0.019] \Rightarrow$  **rules win by parsimony**. **TOP** — rules win outright.
- **Read the random row.** 43 “captures” bought with 508 false alarms. Capture without precision is spraying, and it is on the board so nobody mistakes recall for skill.

# Walk-forward: how the numbers are kept honest

---

**Fit on the past, score the next unseen year, roll forward.** Every figure in this deck is out-of-sample by construction; nothing is scored on data used to choose it.

**Thresholds are learned per year from prior years only:**

2018: 0.886    2020: 0.888    2022: 0.891  
2024: 0.894    2026: 0.895

Stable to the third decimal across nine years — the detector is not re-fitted into a new personality each year.

## Where the false alarms actually land

|                                  |     |
|----------------------------------|-----|
| isolated (no episode $\pm 90$ d) | 143 |
| early (trough within 90d)        | 14  |
| post-peak (within 90d after)     | 12  |

26 of 169 “false” alarms are near-misses on real episodes. They are still counted as false — the anatomy is shown so the failure mode is understood, not so the number can be discounted.

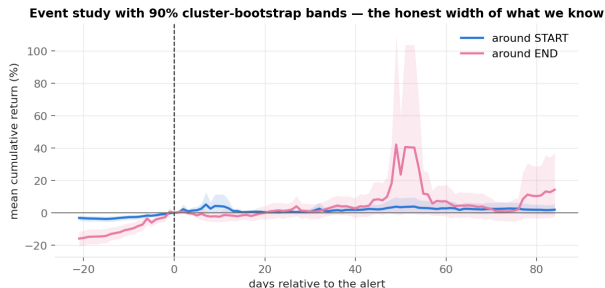
## Fixing the one-day blip — at the signal, not with a rule

| Smoothing            | OUT captured | OUT FA    | OUT AP       | IN captured | IN late   | IN FA      |
|----------------------|--------------|-----------|--------------|-------------|-----------|------------|
| 1 d                  | 26           | 41        | 0.435        | 24          | 12        | 166        |
| 3 d                  | 24           | 41        | 0.438        | 19          | 14        | 146        |
| 5 d                  | 24           | 41        | 0.443        | 19          | 10        | 132        |
| <b>7 d (adopted)</b> | <b>24</b>    | <b>39</b> | <b>0.449</b> | <b>20</b>   | <b>10</b> | <b>124</b> |
| 10 d                 | 25           | 42        | 0.461        | 19          | 12        | 112        |
| 14 d                 | 24           | 41        | 0.476        | 16          | 11        | 112        |

- **The complaint.** A signal that fires for one day, or a START landing days before an END, destroys PM trust.
- **The wrong fix** would be a hard rule — “ignore signals shorter than N days”. That is a band-aid over a signal that is still wrong underneath.
- **The fix taken** is to trigger on the 7-day-smoothed score and to make a day that already satisfies every END gate ineligible to host a START. Result: START-next-to-END adjacency falls **20** → **2**, late starts 21 → 10, false alarms 169 → 124.

**SO WHAT** The recorded cost is real and carried openly: captures fall 29 → 20. The desk decision was that adjacency is the binding constraint, and that trade is written into the parameter register.

# Event study — what happens after a GET OUT



Forward price paths around END alerts against a same-instrument candidate-day baseline, with bootstrap bands. Bands, not point estimates, because at  $n = 67$  a point estimate is a story.

**SO WHAT** At 10 days the END edge is  $-3.3\%$  versus baseline with the interval *excluding* zero — the only horizon in the study where it does. At 3, 21 and 84 days it does not. One significant horizon out of four is a modest, honest result, not a trading system.

## The companion study: can we spot the useful voices early?

**The question** (ported from Chan 2026, ch. 5–7):  
can high-predictive authors be identified from  
*behaviour and reply-graph position* before their  
track record exists?

**The graph as measured:**

12,528 nodes • 38,201 edges • 5,071 labelled • 237  
positives (prevalence 4.7%).

Eight architectures were ported — logistic, MLP,  
label propagation, GCN, GraphSAGE, H2GCN,  
MixHop — in pure NumPy, no NetworkX.

0.098

AP of the shipped  
model

0.047

random floor

0.005

permutation  $p$  (200 per-  
mutations)

0.095

homophily on the class  
we care about

**SO WHAT** The signal is real ( $p = 0.005$ , and a DICE attack that *doubles* the score confirms the edges matter). But homophily on the positive class is 0.095 against 0.963 on the negative — message-passing has almost nothing to pass, which predicted the graph models' underperformance *in advance*.

# The influence result we are not allowed to claim

---

On a **tenure split** — train on established authors, test on newcomers, which is the deployment question — the shipped model earns **AP 0.046 against a 0.050 floor**, AUROC 0.469, and finds **zero** true positives out of 49.

The best row on that split (MLP, AP 0.073) sits above the floor, but a literally-random scorer scores 0.050 on the same 49 positives.

**Read the best row as noise, not as partial success.**

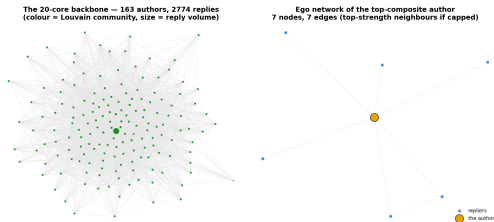
## So what ships?

The influence tab ships as **information only** — a forward-scored, volatility-judged track record per author, with the thesis's “loud but wrong” finding operationalised so the board never ranks by audience size.

It does **not** enter the euphoria detector.

**And note what is *not* the excuse.** The pre-stated maturity bar —  $\geq 130$  labelled positives, set to match the thesis's own scale — *is* met: 237. The store is big enough. The model still does not generalise to new authors, so the blocker is the finding, not the sample size.

# The reply graph, as it actually looks



*Left:* the 20-core backbone — 163 authors, 2,774 replies, node size = reply volume, colour = Louvain community (modularity  $Q = 0.571$  over 3,666 communities). *Right:* the ego network of the single highest-composite author, 7 replying neighbours.

**SO WHAT** Plotting all 12,528 nodes produces a hairball that says nothing; the  $k$ -core is the standard way to show *who talks to whom* in a graph this sparse (mean degree 6.1, median 2, 3,563 isolated nodes). The right-hand panel is the point: even the most central author in the whole store has a *seven-person* audience in reply terms. Influence here is not broadcast, which is why audience-size ranking fails.

## Limitations — pre-answered

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1. **Coverage deserts.** 2023–25 archive coverage is thin to zero. The detectors are blind there, not validated there.
2. **Trough truncation.** The trough is a measured 120-day low, so run lengths cap at 120 days by construction.
3. **V-recoveries qualify.** A boom off a 120-day low admits post-crash rebounds (energy 2020) as episodes — inherited knowingly; the crowd features must *earn* the mania-vs-recovery distinction.
4. **Onset false alarms are above budget:** 0.348 vs 0.23 per instrument-year. Printed on the pane, not excused.
5. **No standalone alpha claim.** Tested, rejected, recorded.
6. **Calibration is not monotone** at low score deciles — annotated on the figure, not smoothed away.
7. **The influence model is transductive.** Its headline numbers assume the author was present when the model was fitted. On unseen authors it falls to the random floor, so it ranks a known board — it does not screen newcomers.
8. **Ground truth requires a confirmed bust.** Euphoria that deflates slowly never enters the catalog, so capture rates speak only to the arcs a desk most needs flagged.

## Future work

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- **Fold comments into the abstracted aggregates** — the single largest capture-recovery lever for the coverage deserts, and the one change most likely to move 2023–25 from “blind” to “measured”.
- **LLM-based call extraction** to replace the VADER stance heuristic, which is the weakest link in the influence study’s labels.
- **Graph features as euphoria inputs** once the influence store passes its maturity bar.
- **Re-admit** `source_breadth` once the multi-source archive spans a full episode cycle — the rejection was about coverage regime, not about the idea.
- **Two watch items auto-re-test at the next research pass:** the logistic-weighted GET OUT (30 vs 24 captures, gain CI  $[-1, +11]$ ) and per-kind thresholds (false alarms halved at equal capture rate, but on unmatched years).

## Conclusion

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- Two rule-based, crowd-only detectors — one for the start of retail euphoria, one for its end — survive a pre-registered tournament against learned challengers.
- They post honest walk-forward records, **fail** an honestly-specified alpha test, and earn their place as **risk timing and monitoring infrastructure** rather than as a black-box signal.
- Every fitted number is either derived from an already-validated constant or walk-forward-learned from the past only. Every negative result is documented with the same care as the positives.

That — more than any single capture rate — is the claim this project defends.

## Appendix — where every number lives

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| Artefact                                | What it holds                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>notebooks/01-07</code>            | The research record. Every chart in this deck is exported from one of them, unmodified.       |
| <code>docs/RESEARCH_REPORT.md</code>    | The living paper; changelog in Appendix C.                                                    |
| <code>docs/PARAMETER_REGISTER.md</code> | Every constant with its class: LEARNED / DERIVED / CONVENTION / GROUND TRUTH / DESK DECISION. |
| <code>docs/DECISIONS.xlsx</code>        | Every adopt/reject decision with its pre-stated criterion and the measured outcome.           |
| <code>docs/research/*.json</code>       | The machine-readable results the notebooks emit; this deck's tables are read from them.       |
| <code>tools/export_figures.py</code>    | Re-exports every figure here from a real notebook run.                                        |
| <code>tools/dashboard_shots.py</code>   | Re-takes every screenshot here from a real dashboard render.                                  |

Deck last edited 2026-07-27 — every frame fitted by measurement rather than by eye — 33 overfull boxes to zero at 77 pages; two silent content losses repaired (the hypotheses slide was pushing its own SO WHAT strip off the page, and `\stat` counters used inside a column were collapsing to 48pt slivers); dashboard screenshots roughly doubled in rendered area by turning those frames sideways and splitting the paired tab slides one-per-slide; long paths made breakable and prose table columns given declared widths

## **Part III — the evidence atlas**

Every remaining exported figure, including the ones that found nothing.

# How to read the atlas

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Three kinds of slide follow, and the labels are literal:

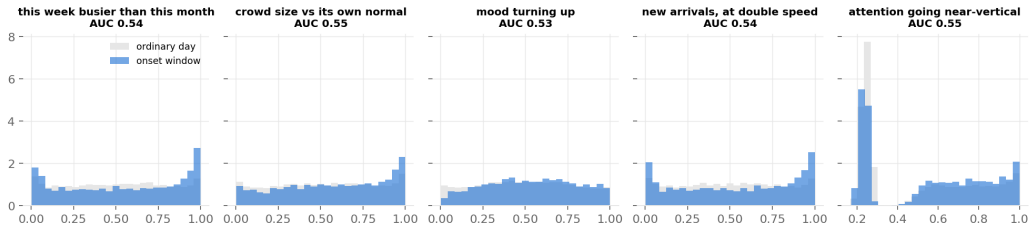
- **DIAGNOSTIC** — a picture of how the shipped thing behaves. No claim is being made; the chart is the audit trail.
- **HONESTY** — a result that makes the system look *worse* and is in the deck anyway, because a reviewer who finds it before you do has won the argument.
- **NULL** — something we tried that did not work. Filed, not hidden. Roughly a third of the research figures are nulls, and that ratio is the point.

## The reproduction path

Nothing here is redrawn. Every PNG comes from `python tools/export_figures.py --all` (notebooks) or `python tools/dashboard_shots.py` (product), and the numbers quoted in the captions come from `docs/research/*.json` emitted by the same runs.

If a caption here and the research record ever disagree, the record is right and this file is stale — which is why the footer of every slide carries the date it was last built.

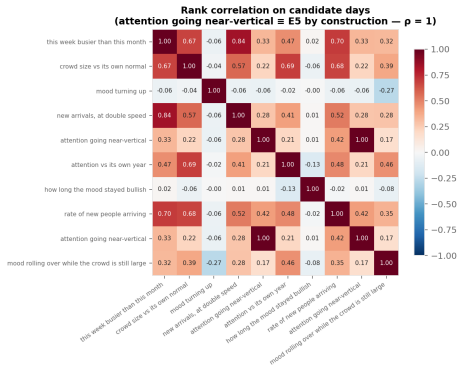
# DIAGNOSTIC — what the features actually look like



Each START-bank feature as two overlaid distributions: grey = an ordinary day, blue = a day inside a 45-day onset window. Per-panel AUCs are 0.54, 0.55, 0.53, 0.54, 0.55.

**SO WHAT** This is what a *weak* feature looks like, and it is worth seeing once. The two distributions almost coincide; the blue mass leans right of the grey only at the very top of the range. Nothing here would survive as a standalone rule, which is the empirical reason the system scores a bank of ten rather than picking a champion.

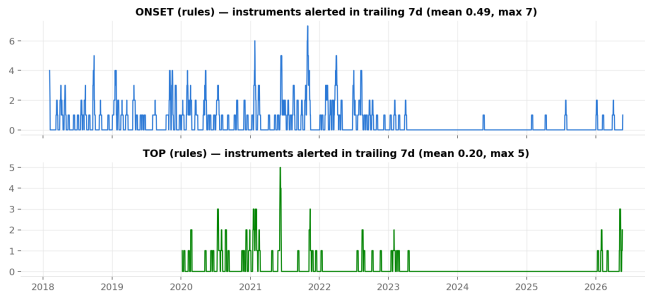
# DIAGNOSTIC — is the bank ten measurements or three?



Spearman correlation between features on candidate days. “Attention going near-vertical” appears twice at  $\rho = 1$  *by construction* — it is the same series entering under two names, and it is printed rather than quietly de-duplicated.

**SO WHAT** Three clusters, honestly. Volume-ish measures are correlated 0.52–0.84 with each other, so the bank has perhaps four independent directions rather than ten. But “*mood turning up*” is nearly orthogonal to everything (0.02 to  $-0.06$ ) and turns  $-0.27$  against “mood rolling over” — so the sentiment channel is carrying information the volume channel cannot. Correlated inputs are why the ablation on the next slides matters; they are not by themselves a reason to cut.

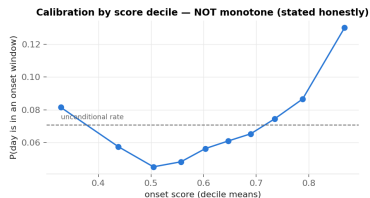
# DIAGNOSTIC — how often does the system actually speak?



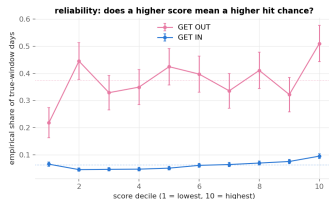
Instruments alerted in the trailing 7 days, over the whole record. START (top): mean 0.49, max 7. TOP (bottom): mean 0.20, max 5, and it starts in 2020 because that is where TOP scoring begins.

**SO WHAT** “Spring” was a design requirement, and this is the receipt. On a universe of 414 priced symbols the detector’s typical week contains *less than one* alert. The 2024–25 thinning is the archive-coverage problem, visible in the alert stream itself — and note that it appears as *silence* rather than as noise, which is the correct failure mode for a system a PM is meant to trust.

# HONESTY — the score is not calibrated, and we say so



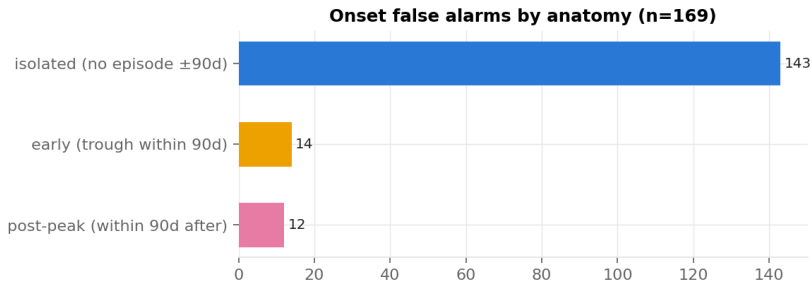
START:  $P(\text{onset window})$  by score decile.  
Non-monotone — the lowest decile (0.082) sits *above* the unconditional rate (0.069) and above deciles 2–6.



The same test at the shipped operating point. GET OUT climbs 0.22 → 0.51 but not monotonically; GET IN is flat near 0.05 until the top decile.

**SO WHAT** The honest reading: the **top decile** is reliably the top decile in both signals, and the middle of the range is not ordered. So the score may be used as a *trigger* and must not be quoted as a probability — “0.70 means a 70% chance” is a claim this chart refutes. That is why the product ships alerts and a state, never a percentage.

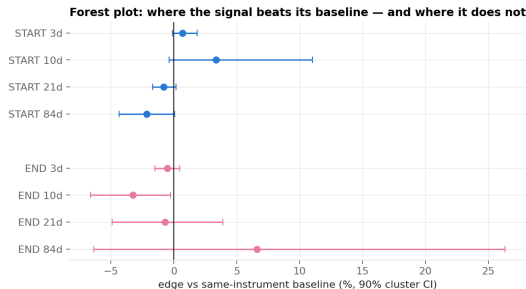
# HONESTY — what the 169 false alarms actually are



The pre-smoothing START false alarms, split by what was happening around them: 143 isolated (no episode within  $\pm 90$  days), 14 early (a trough followed within 90 days), 12 post-peak.

**SO WHAT** 26 of 169 are near-misses on real episodes rather than pure noise, and **all 169 are still counted as false**. The split is shown because it changes what you do next — 143 isolated alarms is a *precision* problem to attack with the score, while 26 timing errors are a *window* problem. It is not shown so the number can be discounted, and it is not netted off anywhere in this deck.

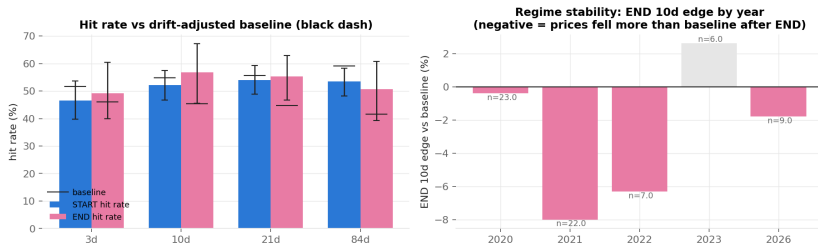
# DIAGNOSTIC — every horizon, both signals, one plot



Edge over a same-instrument candidate-day baseline, in percentage points, with 90% instrument-cluster intervals. Four START horizons above, four END horizons below.

**SO WHAT** Seven of the eight intervals contain zero. The exception is **END at 10 days**:  $-3.3\text{pp}$  with the interval  $[-6.6, -0.3]$  excluding zero. This is the single most defensible slide in the deck *because* it is mostly negative space — a study that found eight significant results out of eight would be a study with a leak in it.

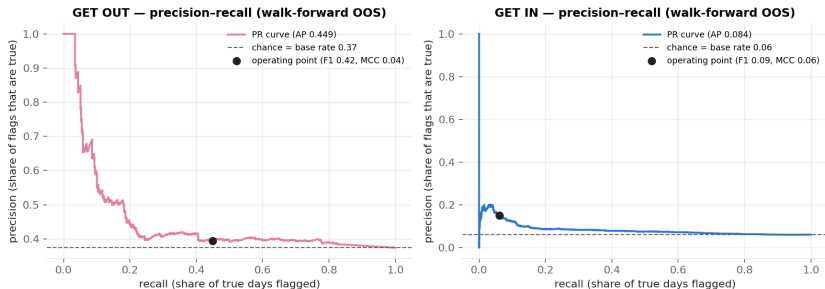
# DIAGNOSTIC — against a drift-adjusted baseline, and by year



*Left:* hit rate at 3/10/21/84 days against the same-instrument baseline. *Right:* the END 10-day edge split by year, with the sample size printed on each bar (negative = prices fell more than baseline after an END).

**SO WHAT** The left panel is the sceptic's control: in a rising market a "price fell" test scores well by accident, so the baseline is the same instruments on non-alert candidate days. The right panel is the regime test — the edge is negative in 2020, 2021, 2022 and 2026, and positive in 2023 on  $n = 6$ . One contrary year on six observations is what an honest small sample looks like; it is greyed rather than dropped.

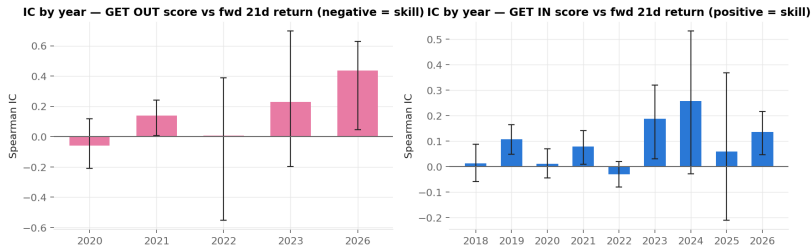
# DIAGNOSTIC — precision–recall at the shipped point



Walk-forward out-of-sample PR curves. GET OUT: AP 0.449 against a 0.37 base rate, operating point  $F_1 = 0.42$ . GET IN: AP 0.084 against 0.06, operating point  $F_1 = 0.09$ .

**SO WHAT** Read the *lift*, not the level. GET OUT's 0.449 looks strong and is only  $1.2\times$  its base rate, because END days are common inside a boom. GET IN's 0.084 looks weak and is  $1.4\times$  its base rate — the better of the two in lift terms. Quoting AP without its floor is the most common way a crowd-data study flatters itself, so both floors are drawn on the chart.

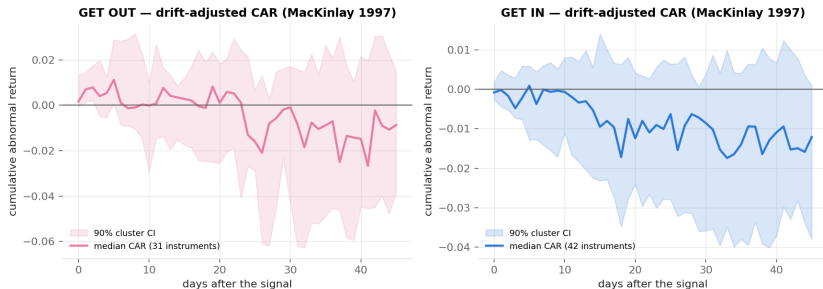
# HONESTY — the information coefficient has a sign problem



Spearman IC by year between score and forward 21-day return. *Left*: GET OUT, where **negative** would mean skill. *Right*: GET IN, where positive would mean skill.

**SO WHAT** The GET OUT panel is the wrong sign in four of five years, and 2026's +0.43 is the worst offender. So a high GET OUT score does *not* rank forward returns downward — it marks the top of a rally that often keeps running. GET IN is right-signed but tiny (0.01–0.26). This is the cleanest single explanation for why both mechanical trading translations were rejected, and it is the reason the product ships a *risk instruction* rather than a return forecast.

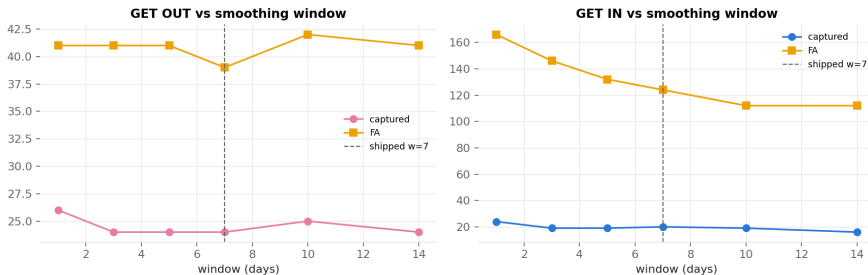
# HONESTY — cumulative abnormal return, MacKinlay (1997)



Median drift-adjusted CAR after each signal with 90% instrument-cluster bands. GET OUT: 31 instruments. GET IN: 42 instruments.

**SO WHAT** Both drift to roughly  $-1\%$  to  $-2\%$  over 45 days and both bands contain zero at nearly every horizon. Note what that means for GET IN: the entry signal is followed by mild *weakness*, not strength. It is a **“the crowd has arrived”** flag, and reading it as a buy instruction is the mistake this chart exists to prevent.

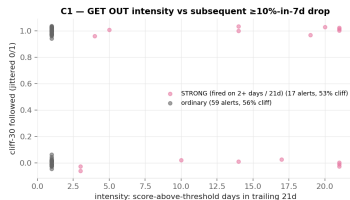
# DIAGNOSTIC — why the smoothing window is seven days



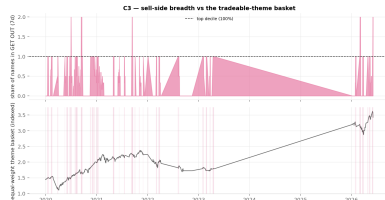
Captures and false alarms against the smoothing window, 1–14 days, with the shipped  $w = 7$  marked.

**SO WHAT**  $w = 7$  is not a round number chosen for tidiness: on the GET OUT panel it is the *minimum* of the false-alarm curve (39) with captures unchanged, and on the GET IN panel it sits where the steep part of the false-alarm decline (163  $\rightarrow$  124) has already been collected while captures are still flat. Past 7 days GET IN captures start falling. The curve is shown so the choice can be checked rather than believed.

# NULL — two refinements that did not earn their place



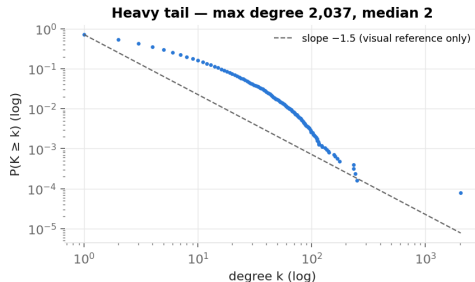
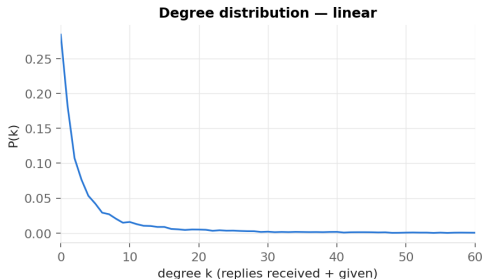
C1: does firing repeatedly help? STRONG alerts (2+ days in 21) are followed by a  $\geq 10\%$ -in-7d drop 53% of the time; ordinary alerts 56%.



C3: does *breadth* of GET OUT across names time the basket? Top decile of breadth marked against the equal-weight theme basket.

**SO WHAT** Both rejected. C1 is the more interesting failure: “it fired louder” carries *no* extra information (53% vs 56%, uplift interval  $[-0.21, +0.22]$ ), which kills the intuitive idea that alert intensity is a conviction measure. These are in the deck because the pre-stated criterion was written before the test, and a criterion only means something if the failures are shown too.

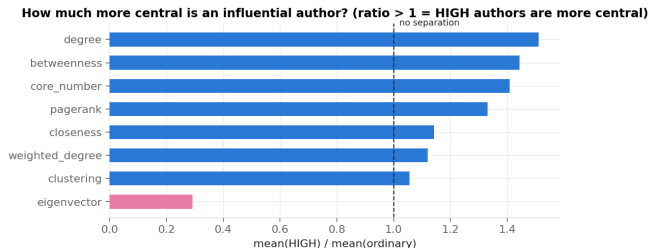
# DIAGNOSTIC — the shape of the reply graph



Degree distribution, linear (left) and as a log-log survival curve (right). Max degree 2,037; median 2; the dashed line is a  $-1.5$  slope for visual reference only, not a fitted power law.

**SO WHAT** Heavy-tailed, as social graphs are. The consequence for the model is concrete: with a median degree of 2 and 3,563 isolated nodes, most authors have almost no neighbourhood for a graph model to aggregate over. That is the structural reason the graph architectures fail later — there is nothing to pass a message across for the typical node.

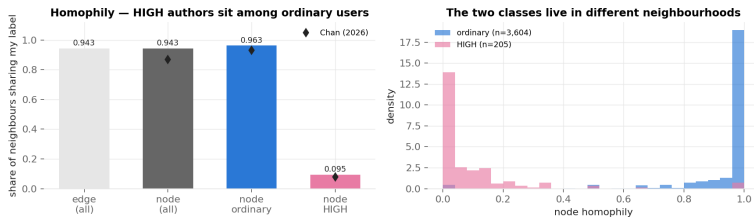
# HONESTY — centrality alone does not find the useful voices



Ratio of mean centrality for high-predictive authors to mean for ordinary authors. Above 1.0 means the useful voices are more central.

**SO WHAT** Degree 1.52, betweenness 1.45, core number 1.41 — real separation, and far too weak to rank on: a  $1.5\times$  ratio on a heavy-tailed distribution means the two groups overlap almost entirely. Eigenvector centrality is *below* 1.0 (0.29), i.e. the useful voices sit *away* from the densest cluster. This directly reproduces the thesis’s “loud but wrong” finding, and it is why the board never ranks by audience size.

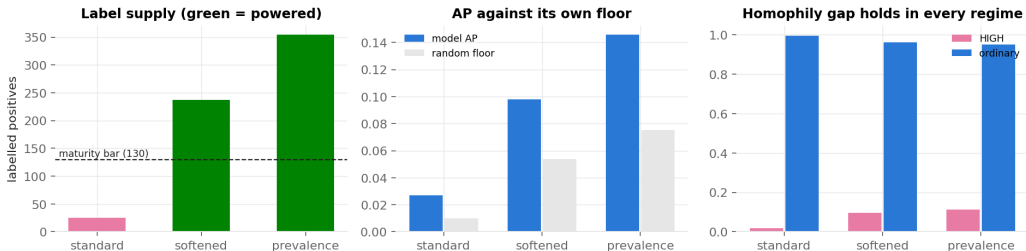
# HONESTY — the useful voices are scattered, not clustered



*Left:* share of neighbours sharing a node's label, with the thesis's own values marked as diamonds. *Right:* the same quantity as a distribution, split by class.

**SO WHAT** Graph-level homophily of 0.943 looks encouraging until it is split: ordinary authors 0.963, high-predictive authors **0.095**. Almost every useful voice is surrounded by ordinary ones. This is the single number that predicts the whole ladder result — message passing works by assuming your neighbours look like you, and for the class we care about that assumption is false. The thesis reports the same pattern, which is reassuring about the method and discouraging about the approach.

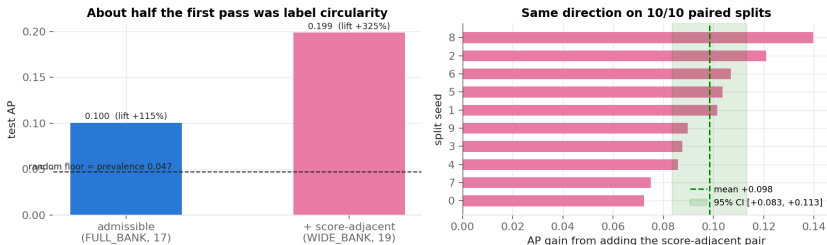
# DIAGNOSTIC — the label regime, and why “softened” ships



Three label definitions against the pre-stated  $\geq 130$ -positive maturity bar (left), model AP against the matching random floor (middle), and the homophily split under each (right).

**SO WHAT** The strict definition yields about 25 positives — below the bar, so it cannot be scored, not because we dislike the answer but because the bar was set in advance. “Softened” yields 237 and ships; “prevalence” yields more still and scores higher (0.148) against a higher floor (0.075), so its *lift* is no better. The right-hand panel is the important one: the homophily gap survives all three, so the finding is not an artefact of where the label line was drawn.

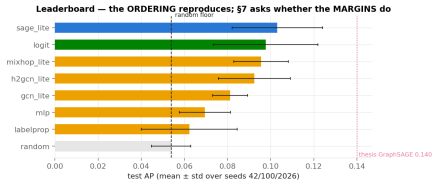
# HONESTY — about half the first pass was circular



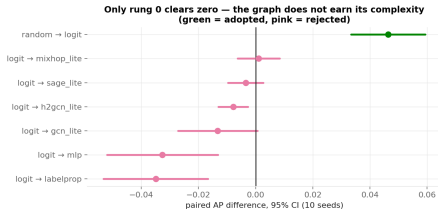
*Left:* test AP with the admissible feature bank (0.100) versus the bank including two score-adjacent features (0.199). *Right:* the AP gain from adding that pair, on each of 10 paired splits.

**SO WHAT** The first version of this study scored 0.199 — twice the honest number — because two features were downstream of the label itself. The gain is +0.098 with a 95% interval of [+0.083, +0.113] and the same sign on 10 of 10 splits, i.e. the leak was large and utterly consistent. Both features are now on a written LEAKAGE blacklist enforced in code. **The headline number in this deck is 0.098, the one after the leak was removed.**

# DIAGNOSTIC — the leaderboard, and the ladder that tests it



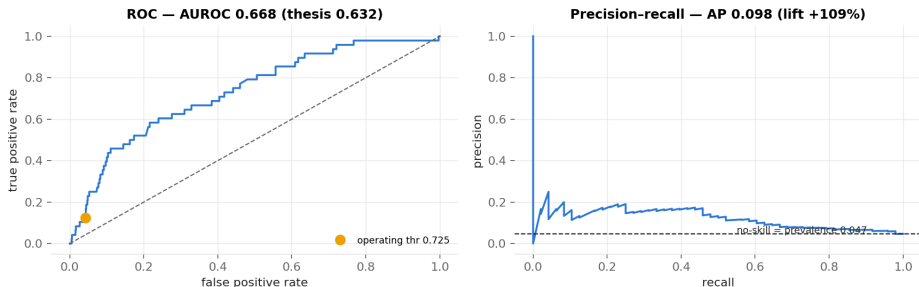
Test AP, mean  $\pm$  std over three seeds. The thesis's GraphSAGE (0.140) is marked; our random floor is 0.054.



The same comparison done properly: *paired* AP differences with 95% intervals over 10 seeds, one rung at a time.

**SO WHAT** The left chart reproduces the thesis's *ordering* — GraphSAGE on top, label propagation near the floor — and would let you claim a graph model won. The right chart is why we do not: only random → logistic clears zero (+0.046, [+0.033, +0.060]). Every graph rung is inside noise or negative. **Overlapping error bars are not a result; a paired test is.**

# DIAGNOSTIC — ROC and PR for the shipped influence model



AUROC 0.668 against the thesis's 0.632 (left) and AP 0.098 against a 0.047 no-skill floor (right). The orange marker is the shipped operating threshold, 0.725.

**SO WHAT** Better than the thesis on both metrics — and this is exactly where a defence gets lost if you stop reading. The operating point sits at about 12% recall: the model finds one useful voice in eight, and is used to *order a board a human then reads*, never to make a call. The next slides are the tests that decide whether even that is safe.

# DIAGNOSTIC — does the result depend on the label recipe?

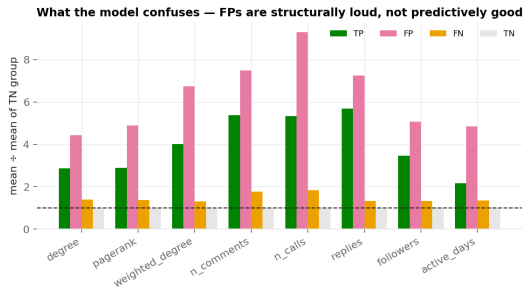
Label-ingredient sensitivity — AP holds even where the recipe reshuffles who is positive



Each point is a different label recipe: test AP against how much of the shipped positive set that recipe also calls positive.

**SO WHAT** Every recipe lands between 0.098 and 0.137, all comfortably above the 0.047 floor, and the *highest* score (0.137, accuracy-only) is the one that agrees with the shipped labels least (74%). Two useful readings: the finding is robust to the recipe, and the shipped 0.4/0.4/0.2 blend is the most *conservative* choice available rather than the best-scoring one. Picking the 0.137 recipe after seeing this chart would be the definition of tuning on the test set.

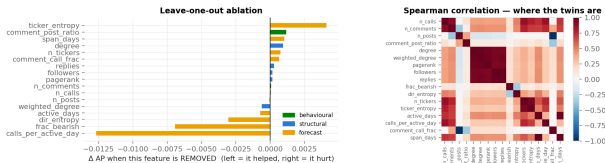
# HONESTY — what the model confuses, and why



Mean feature value of each prediction group, divided by the mean of the true negatives. Bars above 1.0 are louder than a typical ordinary author.

**SO WHAT** False positives are *louder than the true positives* on every single feature — most starkly on call count ( $9.3\times$  versus  $5.4\times$ ). So the model's errors are not random: it is being fooled by volume. That is the thesis's “loud but wrong” author, isolated as a named failure mode, and it is the reason the leaderboard is capped by forward accuracy rather than by activity.

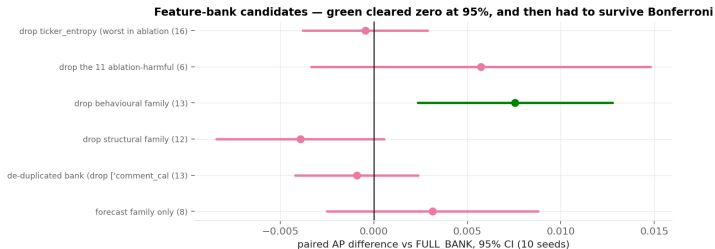
# DIAGNOSTIC — leave-one-out ablation, and where the twins are



*Left:* change in AP when each feature is removed — left of zero means the feature was helping. Colour marks the family: behavioural, structural, forecast. *Right:* the Spearman matrix that shows which features are near-duplicates.

**SO WHAT** Calls per active day, bearish share and directional entropy carry the model; ticker entropy actively *hurts* it. Read the two panels together, though: the structural block (degree, weighted degree, PageRank, followers, replies) is internally correlated at 0.8+, so their small individual ablations mean “the others cover for me”, not “I am useless”. Single-feature ablation on correlated inputs understates every member of a redundant group — which is why the next slide tests whole families instead.

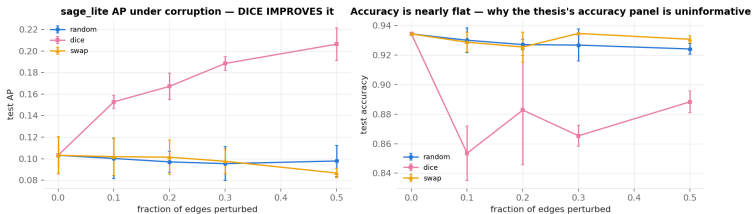
# DIAGNOSTIC — the one bank change that cleared zero



Six candidate feature banks as paired AP differences against the full bank, 95% intervals over 10 seeds. Green cleared zero; pink did not.

**SO WHAT** Only *drop the behavioural family* cleared zero (+0.0075, [+0.0022, +0.0128]) — and it then had to survive a Bonferroni correction across the six candidates tested in the same round, because six shots at 95% confidence gives roughly a one-in-four chance of a false winner. Testing families rather than single features is what made a real effect visible at all; correcting for the six is what stops it being an accident.

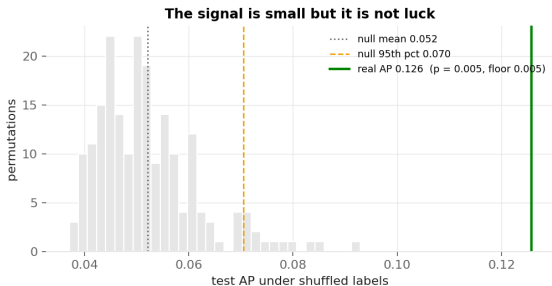
# NULL — when corruption *improves* the model



GraphSAGE-lite under three edge perturbations. *Left*: AP. Under DICE rewiring AP *rises* from 0.10 to 0.205 as half the edges are destroyed. *Right*: accuracy over the same sweep.

**SO WHAT** This is a diagnostic disguised as an absurdity. If deliberately corrupting the graph doubles average precision, the graph structure was not helping — the model was being *hurt* by aggregating over dissimilar neighbours, exactly as the 0.095 homophily predicted, and DICE removes some of them. The right panel also shows why the thesis's accuracy panel is uninformative here: accuracy sits at 0.92–0.94 throughout, because 95% of authors are negative and predicting “no” is nearly always right.

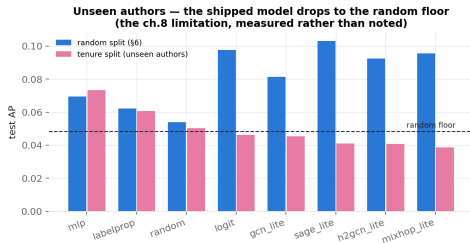
# DIAGNOSTIC — small, but not luck



200 label permutations. Null mean AP 0.052, null 95th percentile 0.070, observed 0.126 on this study's own split ( $p = 0.005$ , the floor for 200 permutations).

**SO WHAT** The observed value sits far outside the shuffled distribution, so the signal is real: chance alone does not produce it. Note the number is **0.126**, not the deck's headline 0.098, because the permutation study uses its own split — quoting it as the headline would be a quiet upgrade. And “real” is not “useful”: the next slide is the test that decides whether it ships.

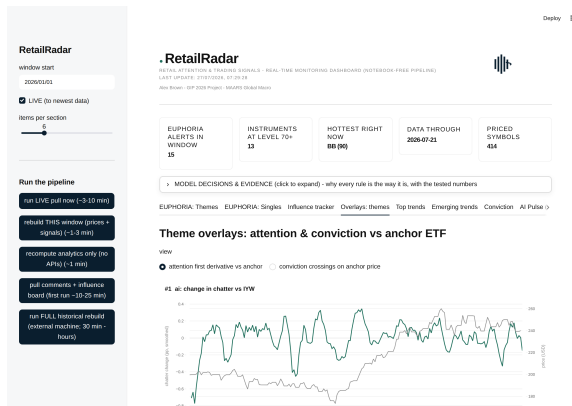
# HONESTY — the test that stopped the influence model



Each architecture twice: on a random split (blue, the number everyone quotes) and on a tenure split where the test authors were never seen during fitting (pink). The random floor is 0.048.

**SO WHAT** On unseen authors every graph model collapses to or below the floor — logistic 0.098 → 0.046, GraphSAGE-lite 0.103 → 0.041 — and the only model that holds is the plain MLP, which ignores the graph and barely clears the floor anyway. The deployment question is always about a newcomer, so this is the chart that decides: the board **ranks a known set of authors** and does not screen new ones. The thesis notes this limitation in chapter 8; here it is measured.

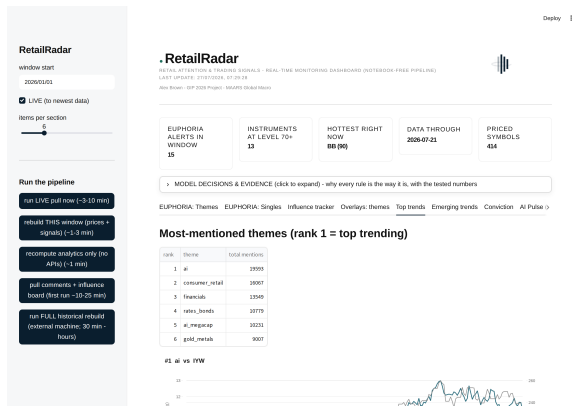
# The product, tab by tab (1/6)



**Overlays.** Change in chatter (green, left axis) against the anchor ETF price (grey, right axis), so a claim about attention can be eyeballed against what price did.

**SO WHAT** This pane is not a signal, and it is deliberately kept. It is where an attention claim gets checked against price by eye before anyone repeats it — the detector's own output lives on the two tabs above, and keeping the raw overlay beside it means a claim can be falsified in a glance rather than taken on trust.

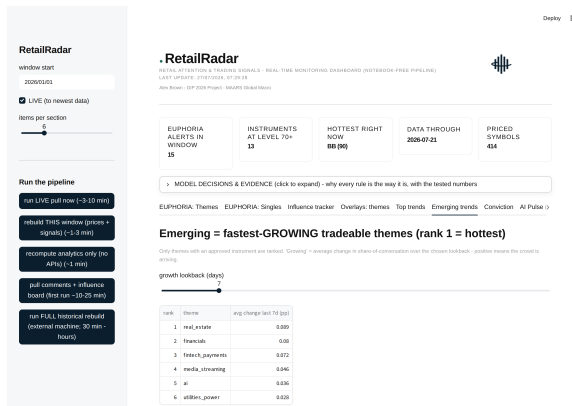
# The product, tab by tab (2/6)



**Top trends.** Most-mentioned themes by raw volume — AI 19,593 mentions, consumer retail 16,067, financials 13,549.

**SO WHAT** Volume ranking is what a PM asks for first (*“what is the crowd loud about right now?”*), and having it on screen next to the detector is what stops anyone mistaking loudness for euphoria — the two lists routinely disagree, and that disagreement is the whole reason the detector is not just a mention counter.

# The product, tab by tab (3/6)



**Emerging.** Ranked by *growth* in share of conversation over a chosen lookback, restricted to themes with an approved instrument — real estate +0.089pp, financials +0.080pp.

**SO WHAT** The filter to tradeable names is the design decision on this pane: growth in share of conversation is only actionable if there is something a desk can actually buy, so an untradeable theme is trivia to a PM and is left off the board rather than ranked and then explained away.

## The product, tab by tab (4/6)

Deploy

## RetailRadar

REAL ATTENTION & TRADING SIGNALS - REAL-TIME MONITORING DASHBOARD (NOTESBOOK-FREE PIPELINE)

LAST UPDATE: 27/07/2028, 07:29:28

Alex Brown - QP 2028 Project - MARS Obedient Music

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☒ LIVE (to newest data)

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items per section  

5

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Run the pipeline

run LIVE pull now (~3-10 min)

rebuild THIS window (prices + signals) (~1-3 min)

recompute analytics only (no APIs) (~1 min)

pull comments ~10-25 min

run FULL historical rebuild (external machine: 30 min - hours)

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### .RetailRadar

|                                 |                                |                              |                            |                       |
|---------------------------------|--------------------------------|------------------------------|----------------------------|-----------------------|
| EUPHORIA ALERTS IN WINDOW<br>15 | INSTRUMENTS AT LEVEL 70+<br>13 | HOTTEST RIGHT NOW<br>BB (90) | DATA THROUGH<br>2026-07-21 | PRICED SYMBOLS<br>414 |
|---------------------------------|--------------------------------|------------------------------|----------------------------|-----------------------|

> MODEL DECISIONS & EVIDENCE (click to expand) - why every rule is the way it is, with the tested numbers

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EUPHORIA: Themes    EUPHORIA: Singles    Influence tracker    Overlays: themes    Top trends    Emerging trends    Conviction    AI Pulse >

## Conviction (rank 1 = most abnormal crowd right now)

> what is conviction? (definition)

Conviction is computed LIVE from the sentiment aggregates (EWMA baseline engine, half-life 40d - validated on real prices with per-year cross-validation) - it can never lag behind a stable file. Grey open triangles on the charts = the signal reverting to neutral, the validated early-exit point.

Negative values are not an error: conviction z is measured against each theme's CWMA trading 84-day normal, so negative = "yes crowd is quieter / more bearish active than it has recently been". After a bull stretch, most themes read negative for a while - that is the mean-reversion of attention, and it is information.

EWMA half-life (days) - smaller = more reactive

10

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| rank | theme           | conviction z (EWMA 10d) | latest z | avg 10d [old metric] |
|------|-----------------|-------------------------|----------|----------------------|
| 1    | europe_defense  | 0.9                     | 1.43     | 0.41                 |
| 2    | energy          | -0.8                    | -1.32    | -1.83                |
| 3    | uranium_nuclear | 0.8                     | 0.27     | 0.4                  |

**Conviction.** How abnormal each crowd is against its *own* trailing 84-day normal, with the EWMA half-life exposed as a control.

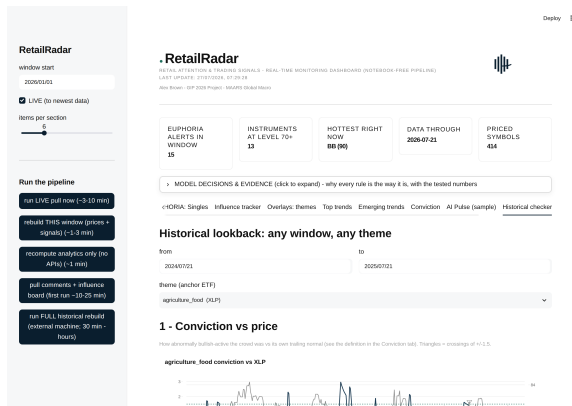
**SO WHAT** This pane explains its own negative numbers on the page — a negative  $z$  means “quieter than this crowd’s own recent normal”, which is information rather than an error. Saying so in the UI prevents the single most common misreading, and exposing the half-life means a sceptic can move it and watch the ranking move rather than take the default on faith.

## RetailRadar • draft v0.3 • last edited 2026-07-27

**AI pulse.** Banner-flagged as preview: the prose is hand-written sample text showing the format a future LLM layer would fill, not output generated from the data.

75

# The product, tab by tab (6/6)



**Historical checker.** Any window, any theme, scored by the same functions the research record uses — so a PM can re-run a claim made on any past date.

**SO WHAT** This is the audit tool, and it is why the deck is falsifiable rather than merely illustrated: every number quoted in Part III can be re-derived by whoever is holding the mouse. It calls the same functions the notebooks call, so a disagreement between this tab and this deck would be a real finding, not a version skew.

# Atlas index — and what is deliberately missing

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**53 exported artefacts; 51 of them appear here or in Parts I–II:**

- Notebook 01 — ground truth (3 figures)
- Notebook 02 — features (5)
- Notebook 03 — rule detectors (2)
- Notebook 04 — tournament and walk-forward (4)
- Notebook 05 — influence (17)
- Notebook 06 — event study and the price gate (5)
- Notebook 07 — performance battery (8)
- Dashboard renders (9)

Re-export: `tools/export_figures.py --all`  
then `tools/dashboard_shots.py`.

## Two figures are held back, and why

06/01 (event study, one vote per instrument) and 05/03 (community sizes) both have *rendering* defects — a legend overlapping a title, and two panel titles colliding.

The results they carry are not in doubt and both appear elsewhere in this deck through other charts. They are excluded because putting an unreadable chart in a defence deck invites the reviewer to wonder what else was not looked at.

Both are on the notebook clean-up list, and the fix is a plotting change plus a re-export — not a re-analysis.